

**The Non-Significant Impact of Profile Attributes on Social Network Centrality
in a MOOC Context: A Comparative Study**

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Abstract

The aim of this study is to understand whether real-world social and professional structures are replicated in online learning communities. By analyzing data from two editions of the MOOC-Ed course “The Digital Learning Transition”, Social Network Analysis (SNA) techniques were applied to measure learners' influence and participation, relating them to their static profiles (Gender, Professional Role and Experience). Contrary to expectations, multiple linear regression (OLS) models yielded statistically non-significant results, demonstrating that professional status or years of experience do not offer any structural advantage within the online network. This “null” result leads to an important pedagogical conclusion: within the MOOC-Ed model, centrality is not hereditary but performative. In other words, reputation is built exclusively through active interaction in the forum, outlining a technically democratic and horizontal environment, although limited by a generally fragmented participation (sparsity).

Keywords: Social Network Analysis, MOOC-Ed, Learning Analytics, Network Centrality, Digital Learning, Online Interaction

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1. Introduction

Massive Open Online Courses (MOOCs) have profoundly transformed the world of education, introducing the possibility of reaching a very large student audience (Fidalgo-Blanco et al., 2015). However, the very nature of MOOCs presents challenges, particularly the extreme heterogeneity of participants, which leads to limited personalized support; this results in high drop-out rates, often between 85% and 95% (Fidalgo-Blanco et al., 2015). The context of this study is the MOOC-Ed (Massively Open Online Course for Educators) model, which aims to provide scalable and accessible professional development to K-12 educators through collaborative learning (Kellogg and Edelmann, 2015a; Kleiman et al., 2013). Specifically, the DLT course (Digital Learning Transition) has a design that embodies key principles of effective professional development, including self-directed and peer-supported learning (Kleiman et al., 2013). Since the effectiveness of this model relies heavily on peer interaction and social dynamics, Social Network Analysis naturally emerges as the ideal methodological tool to measure a learner's engagement and centrality (prestige, influence) within the network (Wise and Cui, 2018; Zou et al., 2021). Although interaction is known to influence learning outcomes, a significant research gap exists concerning the predictive power of participants' static profile characteristics (such as Gender and Professional Role) on the centrality actually achieved. In response to this, the present study examines this relationship across two consecutive editions (DLT1: Spring 2013, DLT2: Fall 2013) (Kellogg and Edelmann, 2015a) of the MOOC-Ed course, posing the following research question:

Given the evolution of the MOOC-Ed DLT course design between DLT1 and DLT2, to what extent do participant profile characteristics (Professional Role, Experience and Gender)

associate with their network dynamics (Participation and Centrality), and how does this association differ between the two MOOC-Ed editions?

2. Related Work

The instructional design of the MOOC-Ed model is founded on the pedagogical concept of social interaction, integrated into the theory of Situated Cognition and Vygotskian thought, according to which the most meaningful learning emerges precisely from active participation within a community of practice (Hung and Chen, 2001). In massive learning environments, peer interaction becomes, therefore, the main vehicle for cognitive and social support (Kellogg et al., 2014), a principle specifically implemented in the MOOC-Ed format to foster professional collaboration (Kleiman et al., 2013). Research previously conducted by the course creators themselves has already explored peer support mechanisms within this specific DLT context (Kellogg et al., 2014). However, their study focused on the comparison between the DLT network and a mathematics MOOC (EQP), leaving the evolution of social dynamics across consecutive editions of the same course unexplored.

To measure these dynamics, Social Network Analysis (SNA) has established itself in the field of Learning Analytics as a designated methodology, making it possible to quantify the learner's position in the structure through centrality metrics (Wise and Cui, 2018). Specifically, this study uses In-Degree as a proxy for prestige and received support, Out-Degree as a measure of participatory initiative, and Betweenness Centrality to identify the 'bridge' role in the information flow (Zou et al., 2021).

The availability of public datasets constitutes today a critical component for the development of effective research pipelines in Educational Data Mining (Mihaescu and Popescu,

2021). However, although the literature has extensively addressed interaction analysis, a research gap remains regarding the predictive association between static profile variables (e.g., Role and Gender) and the dynamic network configurations actually achieved. This investigation is indispensable, considering the non-linear complexity of metrics like Betweenness (Goh et al., 2003), to understand if and how the demographic and professional background influences the acquisition of centrality in the different editions of the course.

3. Dataset Description

This study utilizes the ‘MOOC-Ed network dataset’, publicly available through the Harvard Dataverse, which documents the communicative interactions within two consecutive editions of the course “The Digital Learning Transition in K-12 Schools” (Kellogg and Edelmann, 2015b). The data refer to two distinct editions of the course: the first offered in Spring 2013 (labeled DLT1) and the second in Fall 2013 (labeled DLT2). Although based on the same course, controlled variations were introduced between the two editions: for example, in DLT2 the duration was extended from 6 to 8 weeks to facilitate project completion, and an initial ‘roll-call’ was added to encourage socialization (Kellogg and Edelmann, 2015a).

Operationally, the dataset is organized into four distinct files (.csv): for each of the two course editions, a Nodes file (describing the participants' demographic and professional characteristics) and an Edgelist file (tracking sender-receiver interactions) are provided. The key independent variables selected for this analysis include: Professional Role (e.g., teacher, media specialist, administrator), Experience Level (years of service in education), and Gender (Kellogg and Edelmann, 2015a).

To ensure the validity of the analysis, a rigorous data cleaning process was applied. First, participants identified as ‘staff’ or ‘facilitators’ were filtered and removed, thus considering only interactions among students (peer-to-peer). Second, rows with missing values in the key variables were excluded; finally, the four files were logically merged in the code to allow for a direct statistical comparison between the two course cohorts.

4. Methodology

To answer the research question, this study adopted a quantitative analytical approach structured in two phases: social network modeling (SNA) and inferential statistical analysis using regression.

4.1 Network Construction and Metrics

For each course edition (DLT1 and DLT2), forum interactions were modeled as a graph, where nodes represent the participants and edges represent the interactions (Kellogg and Edelmann, 2015a). Three centrality metrics were calculated on this topological structure, adopting the standard definitions for Learning Analytics described by Saqr and López-Pernas (2024):

1. In-Degree Centrality: calculated as the number of incoming links (replies received). In online learning contexts, this metric indicates social prestige and peer support received (Zou et al., 2021)
2. Out-Degree Centrality: calculated as the number of outgoing links, i.e., replies sent. It represents participatory activity and individual effort in contributing to knowledge construction (Wise and Cui, 2018)

3. **Betweenness Centrality:** quantifies the frequency with which a node acts as an intermediary on the shortest path between two other nodes. It identifies participants who act as bridges in the network's information flow (Goh et al., 2003)

4.2 Statistical Modeling

We used Multiple Linear Regression (OLS - Ordinary Least Squares) to test the impact of profiles on centrality. OLS was chosen because it allows for estimating the specific effect of each predictor on the target variable, providing interpretable coefficients (Burton, 2021). Three models were estimated, one for each centrality index, using the following structure:

$$\text{Centrality} \sim \text{Gender} * \text{Edition} + \text{Role} * \text{Edition} + \text{Experience} * \text{Edition}$$

The analysis focuses on verifying whether the coefficients are significant ($p < 0.05$), to determine if static profiles significantly influence the position within the network.

5. Results

This section presents the results of the sequential analysis, proceeding from the descriptive examination of the network to the inferential verification of the research hypotheses through regression models.

5.1 Descriptive and Visual Analysis

Descriptive statistics reveal that both networks (DLT1 and DLT2) are extremely sparse, a characteristic consistent with the literature on large-scale online courses (Kellogg and Edelmann, 2015a). The median for In-Degree and Out-Degree in both editions is about 1.0, indicating that most participants interacted only once or very few times. In fact, the ratio between the

maximums and the median confirms the presence of few dominant hubs, creating a long-tail distribution (Wise and Cui, 2018).

Visual inspection of the graphs allows us to better qualify these structural dynamics. Specifically, the comparison by Professional Role (Figure 1) highlights that, although some categories like Librarian/Media Specialists show peaks of activity in the first edition, a systematic dominance of one profession over the other does not emerge; the wide error bars suggest instead that engagement is a variable driven by individual motivation rather than by job function.

This "bottom-up" hierarchical distribution is further confirmed by the shape of the Violin Plots related to Betweenness Centrality (Figure 2): the extreme width of the base indicates that for the vast majority of participants (regardless of gender) the intermediation power is null. However, in DLT2, a slight qualitative change is noted in the female distribution, which appears less dispersed at the base and more projected upwards, suggesting a consolidation of brokerage in a restricted elite.

Finally, the most critical evolution is revealed by the correlation Heatmaps (Figure 3): while reciprocity between sent and received messages remains constant and high, in the second edition a significant increase in the correlation between Relevance (In-Degree) and Brokerage (Betweenness) is observed. This signals a progressive "convergence of power" in DLT2, where the most popular actors have become increasingly coincident with those who control the information flow, outlining a network structure where different forms of social capital tend to accumulate in the same nodes.

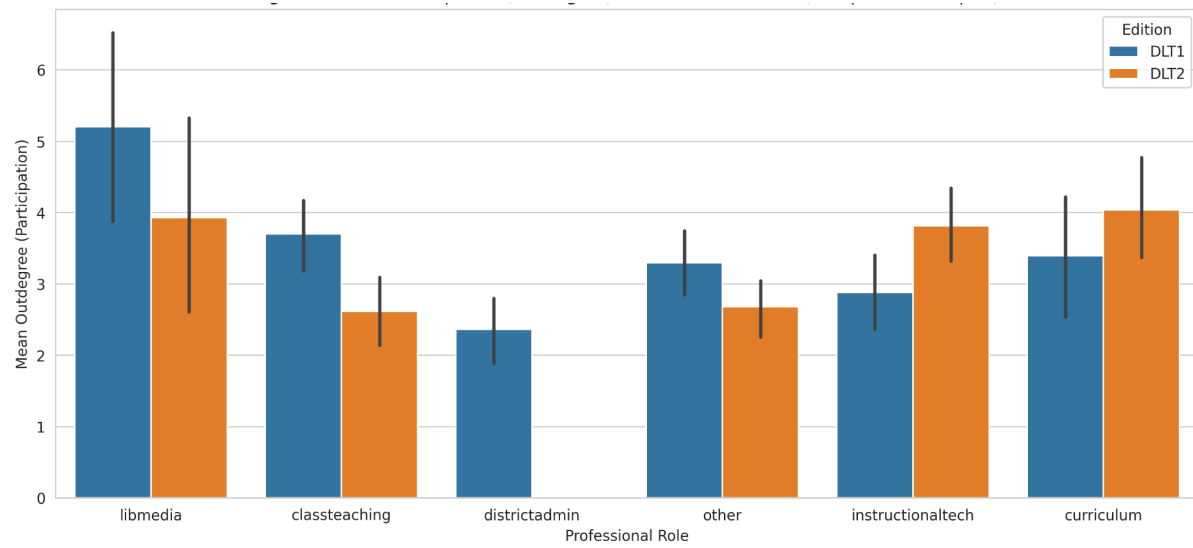
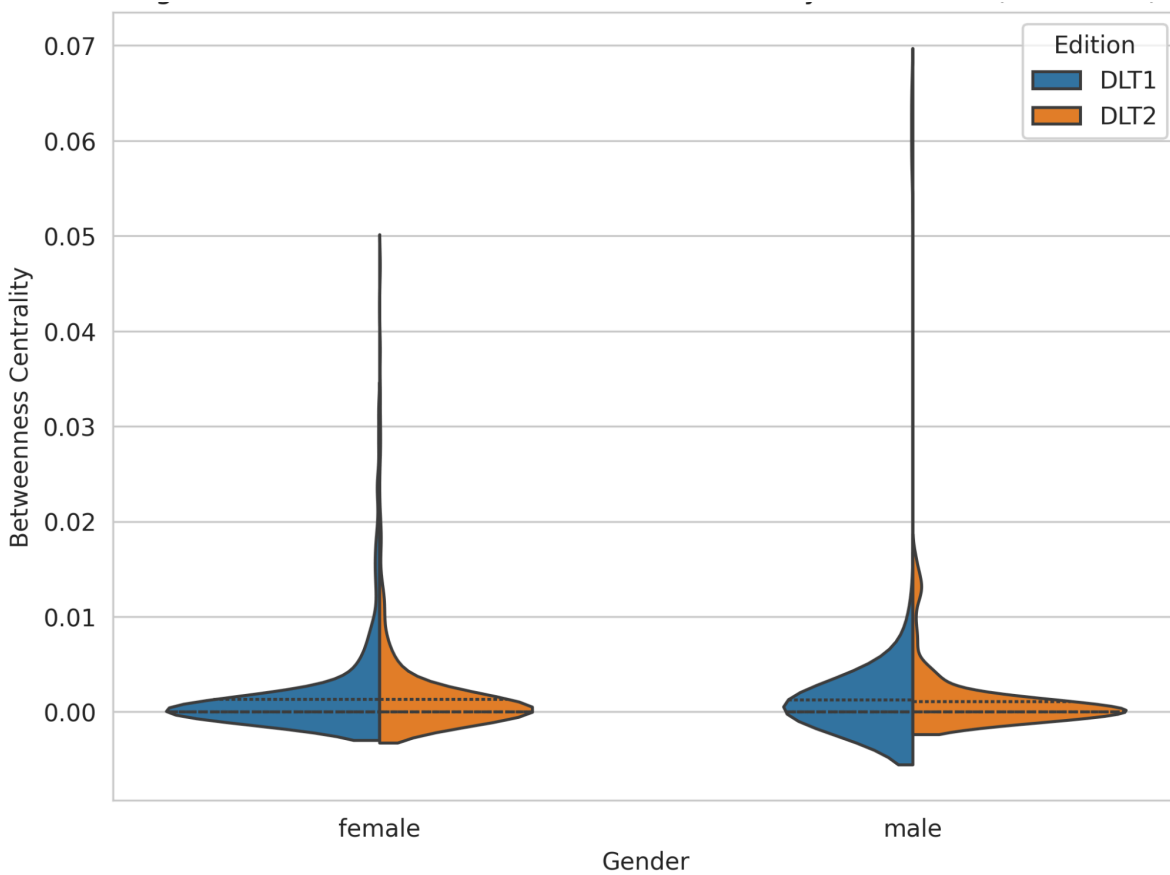
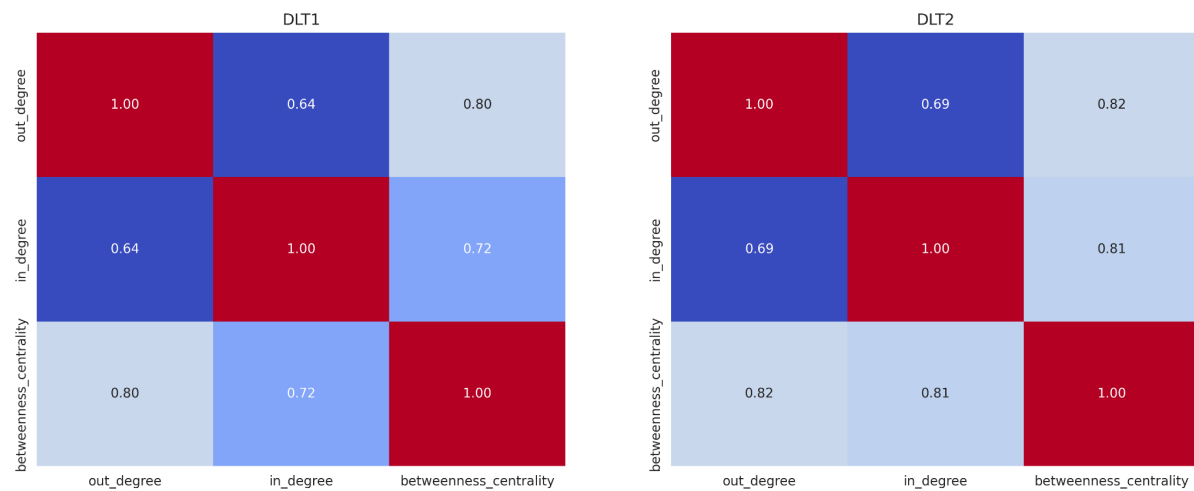
Figure 1*Mean Participation (Outdegree) vs. Professional Role (Comparative barplot)***Figure 2***Distribution of Betweenness Centrality vs. Gender (Violin Plot)*

Figure 3*Correlation Heatmap of SNA metrics for DLT1 and DLT2 (Comparison)*

5.2 Inferential Analysis (OLS Regression)

To rigorously answer the research question regarding the predictive power of profile characteristics, three OLS regression models were estimated. The results are summarized below (Table 1).

Table 1

Model	Dependent variable	R ² adjusted	Prob (F-statistic)	Statistically significant result
Model 1	In-Degree Centrality	≈ 0,000	0,442	NO
Model 2	Out-Degree Centrality	≈ 0,002	0,330	NO
Model 3	Betweenness Centrality	≈ -0,001	0,507	NO

Summary of OLS regression results (DLT1 vs. DLT2)

Following the guidelines for regression analysis (Burton, 2021), we examined both the model fit and the individual coefficients:

- Explanatory Power (Adjusted R-squared): in all estimated models, it is extremely close to zero. This indicates that the static profile variables considered (gender, professional role, experience level) explain practically no variance in network centrality;
- Overall Significance (F-statistic): the F-statistic is not significant ($p > 0.05$) for any model. This implies that, as a whole, the profile variables do not create a significant statistical model to predict a learner's centrality compared to the simple mean (Burton, 2021);
- Coefficients (P-Value): no single regression coefficient, including the Gender:Edition interaction terms, reached the significance threshold of $P < 0.05$;

Based on this statistical evidence, the answer to the RQ is that there is no statistical association between the participants' static profile characteristics and their network dynamics. Furthermore, the evolution of the course design between DLT1 and DLT2 did not change this neutrality. This suggests that, in the MOOC-Ed network, status is not determined by professional background or demographic characteristics, but is likely driven by endogenous behaviors (such as participation frequency) rather than exogenous traits.

6. Discussion

The results of this study show a counter-intuitive but pedagogically relevant perspective on social dynamics within MOOC-Ed courses. Indeed, contrary to the idea that traditional hierarchies (e.g., Principals vs. Teachers) or demographic factors influence the online network structure, the regression analysis revealed a structural neutrality.

6.1 The Democratization of Participation

The absence of statistical significance indicates that status within the DLT network is not influenced by the participant's background. This result aligns with the design goals of the MOOC-Ed model described by Kleiman et al. (2013), which aimed to promote collaborative learning based on horizontal exchange rather than vertical transmission. This suggests that identity within the virtual community is constructed through individual interaction, confirming the view of Hung and Chen (2001) that belonging is defined by active participation and not by a priori attributes. Thus, for example, a novice teacher has the same chance of becoming a central hub as an experienced administrator, given equal participation in the network. This reflects the transformative potential of online learning, capable of dismantling the hierarchical and monolithic structure of traditional schooling in favor of a more flexible and student-centered system (Horn and Staker, 2011).

Such neutrality, especially regarding the level of experience, is also found in the ERGM (Exponential Random Graph Models) analysis conducted by Kellogg et al. (2014) on the same course, who attributed the absence of a 'mentoring effect' (where novices seek out experts) to the difficulty participants found in identifying the real skills of their peers within the platform. Furthermore, recent studies on MOOCs for educators suggest that engagement is primarily driven by the immediate relevance of content for teachers' practical needs ('just-in-time professional development'), a factor that prevails over demographic or prior characteristics of participants (Boltz et al., 2021).

This ‘equity’, however, clashes with the problem of general low participation (sparsity). The network suffers, in fact, from a "winner-take-all" distribution (Wise and Cui 2018), in which a very small minority sustains the entire load of the discussion.

6.2 Limitations

This study, based solely on log metadata (who replies to whom), does not examine metrics on the semantic quality of the contribution or on learning outcomes, factors that Wise and Cui (2018) indicate as essential to distinguish superficial interaction (e.g., simple greetings) from deep learning. In fact, high centrality does not necessarily imply high cognitive quality. Furthermore, the exclusion of facilitators, although necessary to isolate peer-to-peer dynamics, might have hidden the ‘trigger’ role that staff exercises in activating less experienced learners.

7. Conclusion

This Data Analytics project explored the dynamics of two learning networks of the Digital Learning Transition MOOC-Ed. We demonstrated that, although descriptive analyses might suggest differences between editions, OLS inferential tests reveal a statistical non-significance of the impact of static profiles (Gender, Role, Experience) on network centrality. This result highlights the ability of the MOOC-Ed model to favor a horizontal interaction structure free from hierarchical barriers based on role or experience (Kleiman et al., 2013). However, the scarcity of connections and the dependence on a few active hubs (Kellogg and Edelman, 2015a) indicate that democratic access does not automatically translate into widespread participation. Future research will, therefore, have to integrate semantic analysis to understand how to transform this passive presence into active learning for the whole community.

Data Availability Statement

The code, datasets, and supplementary materials for this study are openly accessible via GitHub at: <https://github.com/annagiacometti/mooc-sna-evolution>

The final version of this record is registered under the DOI:

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